0.3 Report Template – Ship Design Project

Ship and Ocean Systems Design 2026 · Instituto Superior Técnico

Group	Baseline	Students (name, number)	Specializations	Date
[Group n.º]	[A Azores Express / B Azores Feeder]	[names and numbers]	[two specializations]	[date]

How to use this template: keep the numbered sections, which follow the ten deliverables D1–D10 of 0.2 Ship Design Project. Replace every text in brackets and delete these italic notes. Hand in the report, the datasheet and your files in one ZIP by e-mail, by 13 December 2026. Declare any use of AI tools in section 0.

0 Summary

Half a page: the ship, its main particulars, the design objective and how well the design meets it. Declare here the AI tools used and what for.

D1 Design Basis

Must show: owner's requirements, port limits, environmental targets. Use the method of Exercise 1.1 (practised on the reference ship) for your own ship, with every requirement tagged and sourced.

Requirement	Value	Unit	Source	Status (given, assumed, checked)
[Capacity]	[ ]	[TEU]	[0.2]	[ ]
[Service speed]	[ ]	[kn]	[ ]	[ ]
[Port draught limit, each port]	[ ]	[m]	[ ]	[ ]

D2 Reference Ship Database

Must show: similar ships under 5 years old, their sources, and the regressions with their bounds.

Ship	Year	TEU	DWT [t]	LOA [m]	LPP [m]	B [m]	D [m]	T [m]	V [kn]	P [kW]	Source
[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

For each regression: the equation, R^2, the standard error and the range of validity. Plot every regression with your ship on it.

D3 Ship Model

Must show: inputs and outputs, the calculation sequence, and a closed weight balance (Δ = lightship + DWT, within your tolerance).

D4 Voyage and Financial Model

Must show: ship model outputs used as inputs, port restrictions, revenue assumptions. Round voyage or loop time, fuel or energy per voyage, costs, required freight rate and NPV.

D5 Validation Note

Must show: the model run for reference ships, the error per quantity, and how the model was corrected.

Quantity	Reference ship	Model	Error [%]	Action
[Lightship]	[ ]	[ ]	[ ]	[ ]

D6 Optimization Report

Must show: a justified objective, the design variables and their bounds, the active constraints, and the optimum compared with the requirements.

D7 Compliance Sheet

Must show: attained EEDI against required and CII rating, or not applicable, with the reason; energy per TEU-mile (kWh/TEU·nm) for the battery-electric Feeder; freeboard and intact stability, draught at every port of call.

Check	Required	Attained	Pass
[EEDI]	[ ]	[ ]	[yes / no / not applicable, with the reason]

Check	Required	Attained	Pass
[Draught at each port]	[ ]	[ ]	[ ]

D8 General Arrangement and Hull Form

Must show: an A3 sketch with profile and midship section, and a hydrostatics check of the hull form against the ship model.

D9 Specialization Documents

One subsection per specialization, as its self-study deck asks: calculation note, required drawings, acceptance checks.

9.1 [Specialization 1]

9.2 [Specialization 2]

D10 Datasheet and Presentation

Attach the one-page A4 datasheet (0.4 Datasheet Template). The 15-minute presentation on 18 December follows the same order as this report.

References

Author (year), title, publisher or journal. Cite Papanikolaou (2026) by section.

Appendices

Code, spreadsheets, and the full reference ship data, as files in the ZIP.